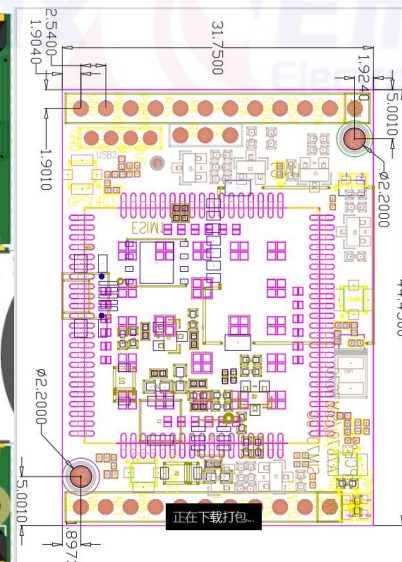
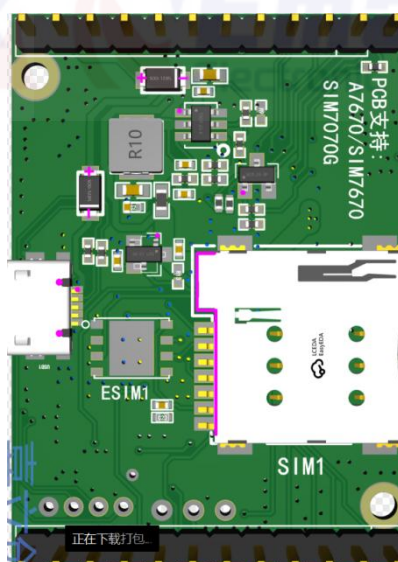
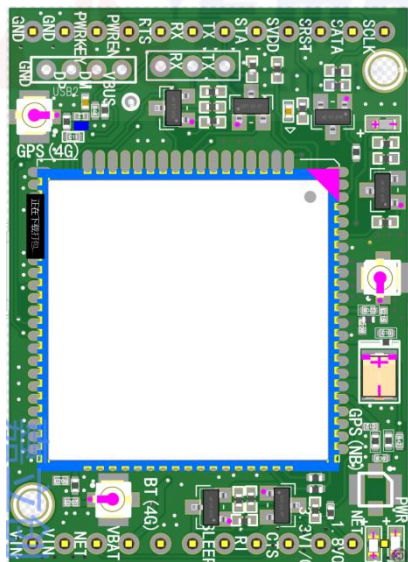


24&24 3.0 Core Board Specifications

制式	型号	制式	型号	制式	型号
GSM	SIM800A-L	LPWA	SIM7070E	LTE	A7670C-FANS
	SIM800A32-L		SIM7070E		A7670C-FASC
	SIM800F-L		SIM7070E-REEL		A7672S-LASE
	SIM800F-L		SIM7070E-MN		A7672S-FASE
LPWA	SIM7000C		SIM7070G-HP		A7670C-GW
	SIM7000C		SIM7070E		A7670G-LLSE
	SIM7000E		SIM7070G-HP		A7670C-LASE-PCIEA
	SIM7000E		SIM7070G-M		A7670C-LASS-PCIEA
	SIM7000A		A7670C-BASL		A7670E-LASE-PCIEA
	SIM7000A		A7670C-BASS		A7670G-LABE
	SIM7070E-NG		A7670E-FASE		A7670C-LANV
	SIM7000JC		A7670E-LASE		A7670C-LANS
	SIM7000JC		A7670E-LASC		A7670C-MANV
	SIM7000G		A7670E-MASC		A7670C-MANS
LPWA	SIM7070G-NG	LTE	A7670SA-FASE		A7670C-LASE
	SIM7070G-NG-REEL		A7670SA-LASE		A7670C-LASC
	SIM7070G		A7670SA-LASC		A7670C-LASL
	SIM7070G-REEL		A7670SA-MASC		A7670C-LASS
	SIM7070G-PCIE		A7670C-FASL		A7670C-MASL
	SIM7070G-MN		A7670C-FASS		A7670C-MASS

★Core Borad Size



PinNumber (P1)	NAME	Pin Description	Remark
1	SC	USIM1 Bus clock output	When the SIM card is external, the external SIM card holder can be directly connected to the SC, SD, SR, SV pins. Note: The onboard card holder and the external card holder cannot be inserted at the same time
2	SD	USIM1 Bus data	
3	SR	USIM1 Bus reset output	
4	SV	USIM1 card power supply output, output voltage can be changed dynamically according to the external card type, output current maximum 50mA	
5	STA	Power-on status indication: when the core board is powered on and initialization is complete, STA outputs a high level, otherwise it maintains a low level	Used to determine whether the core board is powered on. Voltage domain: 3.3V
6	TXD	Main serial port sending, used for AT command configuration and data transmission	Voltage domain: 3.3V; other levels can be changed by enabling VTTL
7	RXD	Main serial port receiving, used for AT command configuration and data transmission	
8	RTS	Hardware flow control, main serial port RTS input.	Voltage domain 3.3V
9	PEN	The core board power enable pin is pulled up to VIN by default. It will automatically start when powered on. When pulled down, the core board power supply has no output. If not used, leave it unconnected.	
10	P/R	Module power on/off/reset control, default is PWK pin	
11	GND	Negative power input	
12	P/G	Default is GND	
Pin Number (P2)	NAME	Illustrate	Remark
13	VEXT	Internal 1.8V power output, maximum output current 50mA, cannot power high power loads	Electrostatically sensitive, no need to be suspended
14	VTTL	External reference level input for level conversion circuit, invalid by default	
15	CTS	Hardware flow control, main serial port CTS output	Voltage domain 3.3V

16	RI	Ringing indication, when receiving URC report, it outputs low level for 60ms, and then returns to high level state	Voltage domain 3.3V
17	DTR	Sleep wake-up: when the core board enters sleep mode, pull down DTR to wake up the core board	Voltage domain 3.3V
18	TXD2	Serial port 2 sends, not used by AT firmware, just leave it floating	Voltage domain 3.3V
19	RXD2	Serial port 2 receiving, not used by AT firmware, just leave it unconnected	Voltage domain 3.3V
20	VBAT	Module power input, supports 3.4~4.2V input, typical value 3.8V, can be used for lithium battery power supply; when using VIN power supply, VBAT output 3.8V	PEN pin is disabled when powered by VBAT
21	NET	Network status indication	
22	VIN	Power input positive, supports 5-16V DC input	You can connect one or both.
23	VIN		
引脚号 (P3)	NAME	说 明	备注
24	TXD	Send through the main serial port, which is convenient for monitoring the data interaction between the main control and the core board after the core board is installed	To assist in testing
25	RXD	Main serial port receiving, convenient for monitoring the data interaction between the main control and the core board after the core board is installed	
26	GND	Main serial port auxiliary test common ground pin	
27	VB	USB_VBUS, VBUS presence detection input	USB XH2.0 Terminal Blocks
28	UN	USB_N, USB bus differential negative	
29	UP	USB_P, USB bus differential positive	
30	GND	GND, ground	

Note:

1. For conventional applications, only VIN, GND, TXD, RXD, and PEN are needed, and other pins can be left floating.
2. VBAT and VIN cannot be used as power input at the same time. When VBAT is used to power the core board, the PEN pin becomes invalid..

Application Interface

1 Power supply

The core board has three power supply modes: VIN, VBAT, and USB.

VIN: The power supply range is 5V~16V, the power needs to be more than 8W, and the recommended power supply is 5V 2A.

VBAT: The power supply range is 3.3V~4.3V, generally used for lithium battery power supply.

USB: Powered by 5V USB data cable. Some computer USB interfaces (it is recommended to connect to computer USB 3.0) have insufficient driving capacity,

The core board may not start or restart frequently. At this time, you need to connect VIN to power the core board.

Note:

1. VIN and VBAT cannot power the core board at the same time.
2. The actual input voltage must be within the minimum and maximum value range.

2 Power on/off

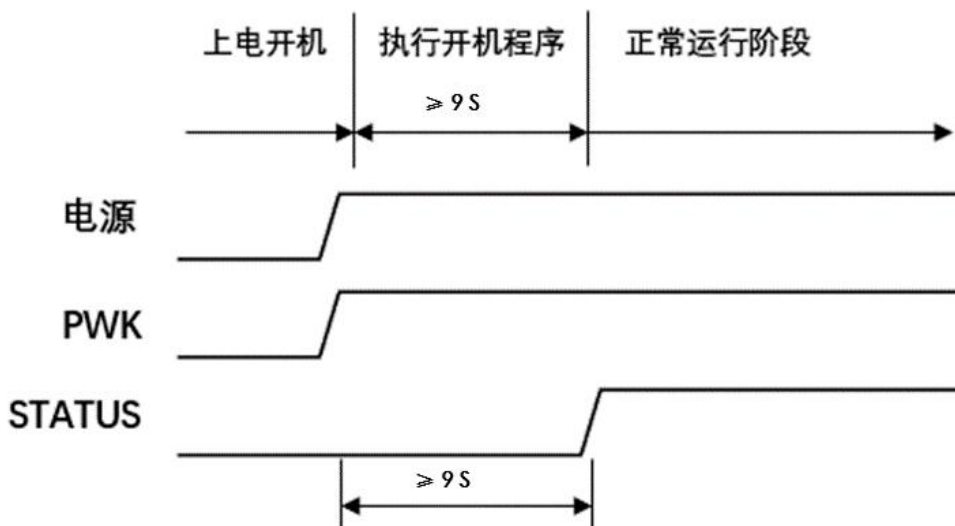
The core board automatically powers on by default. In applications, it is recommended to use the PWK and PEN pins to control the power on and off of the core board in order to reduce power consumption and handle abnormal situations.

应用场景	PWK	PEN
设备长连接, 无开关机需求	可悬空	建议启用, 当遇到异常时, 控制 PEN 强制断电重启
有低功耗需求/不需要长连接的应用, 需要控制核心板开关机	启用, 通过主控 IO 控制核心板开关机	建议启用, 但 PWK 失效时, 控制 PEN 强制断电重启

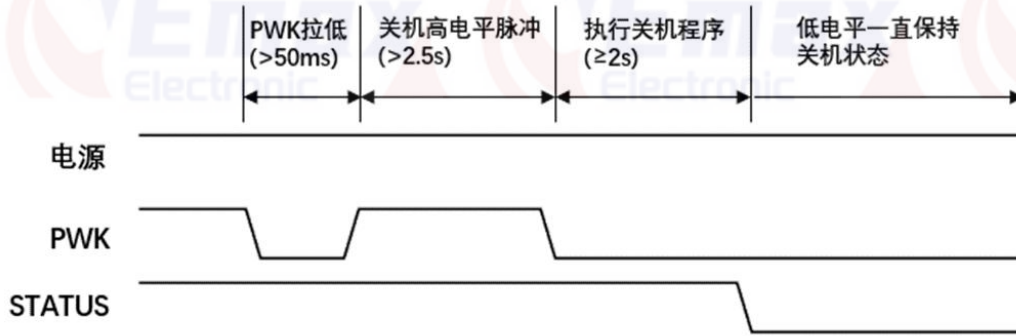
Note: Under normal circumstances, use the PWK pin to shut down. Only when PWK fails, pull down the PEN pin to disconnect the core board power supply to shut down. It is not recommended to shut down through the PEN pin frequently, because long-term abnormal operation may damage the file system in the Flash, causing the core board to fail to boot normally.

2.1 PWK

Power on: The core board automatically powers on by default. The power on sequence is shown in the PWK power on sequence diagram:



Shutdown: In the power-on state, pull the PWK pin low for more than 50ms and then pull it high for at least 650ms to enter the shutdown procedure. At this time, the PWK pin is pulled low and maintained for more than 2S to ensure that the shutdown procedure is fully executed (after the shutdown procedure is executed, the core board will remain in the shutdown state if the low level is maintained). The shutdown timing is shown in the PWK shutdown timing diagram:



Note:

When the PWK pin fails to shut down, you can control PEN to directly cut off the power supply to shut down;

The voltage domain of the PWK pin is 3.3V.

2.2 PEN

When the PWK pin fails to shut down, you can control PEN to directly cut off the power to shut down. When using the MCU's GPIO to control the PEN

pin, it is recommended to add a transistor driver.

Power on: By default, PEN is internally pulled up to VIN, that is, the core board is enabled by default, and it is powered on.

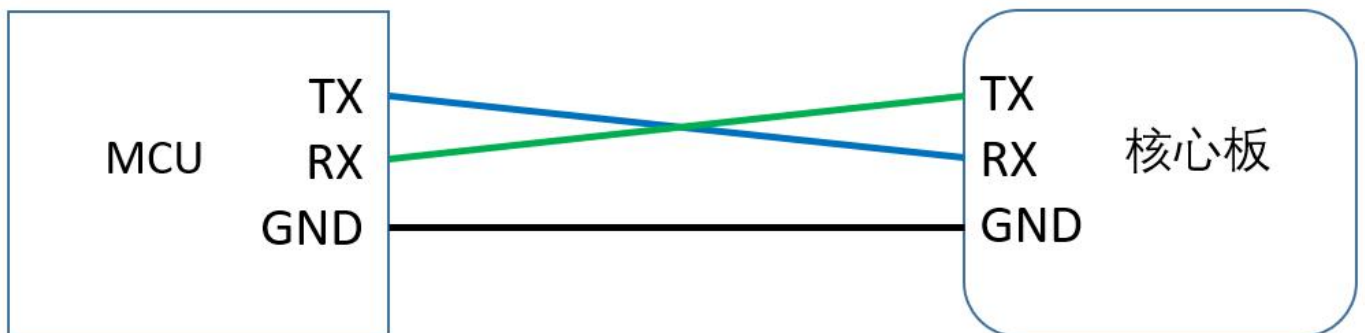
Power off: Pulling down the PEN pin disconnects the power supply of the core board to achieve power off and shutdown.

3 TTL

The core board has two serial ports, serial port 1 and serial port 2. Serial port 1 is the main serial port for AT command interaction and data transmission. Serial port 2 has no function in AT firmware and can be left floating.

Supported baud rates are: 4800, 9600, 19200, 38400, 57600, 115200, 230400, 460800bps, etc.

When communicating with the master control, TXD and RXD are cross-wired.



The default serial port level is 3.3V TTL. If the user's main control serial port level is not 3.3V, level matching is required. You can modify the jumper resistor on the core board and connect the VTTL pin to the target voltage.

IO 电平值	更改说明	备注
3.3V	焊接 RT1, 不焊接 RT2	板子出货默认焊接方式
其他电压	焊接 RT2, 不焊接 RT1, VTTL 接到目标电压	需要自行更改或联系我司销售

Note:

After modifying the IO level, the levels of the TXD, RXD, TXD2, RXD2, DTR, RI, CTS, RTS, and STA pins will all become VTTL.

3.1 DTR wake-up

The DTR pin is the sleep mode trigger pin of the core board. It can be left floating when not in use. Entering the sleep mode can significantly reduce the standby power consumption. In the sleep mode, the main serial port function will be turned off. When the host and the core board are connected through the serial port, the module can be put into sleep mode by the following steps: Use the AT+CSCLK=1 command to enable the sleep function. Make sure DTR is kept at a high level (the default internal pull-up is high).

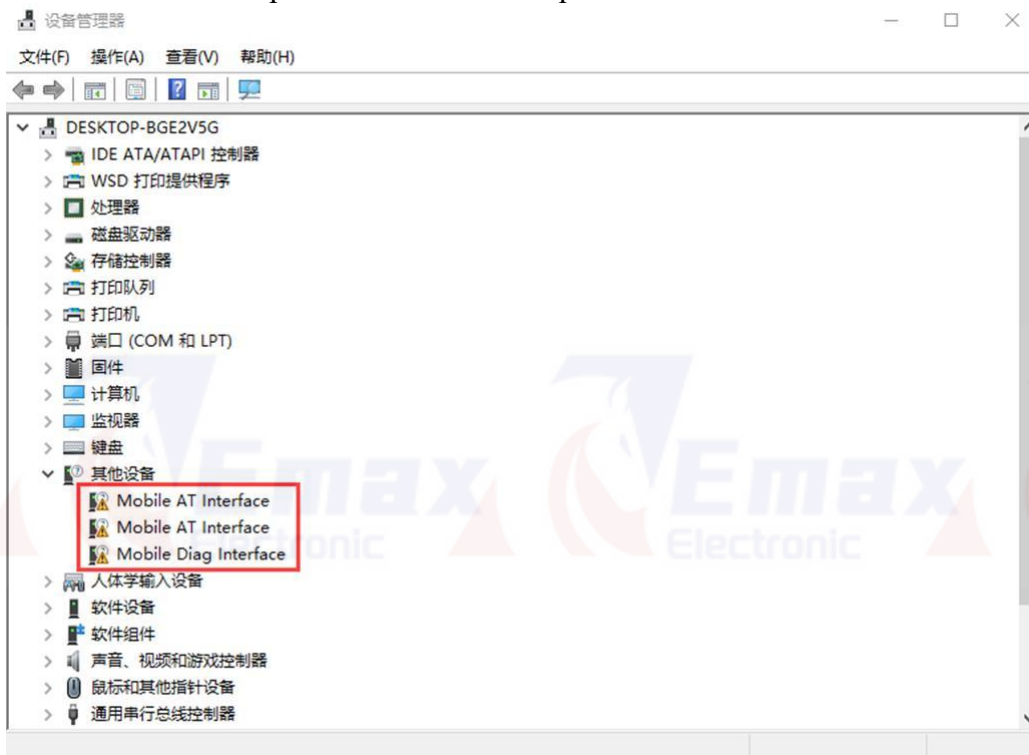
The module can be woken up by pulling DTR low by the host (it needs to be kept low in sleep mode to keep the core board awake).

4 USB interface

Note:

1. The USB cable must have data transmission function.
2. USB driver needs to be installed to use USB interface communication.

After the core board is connected to the computer via USB, if the driver is not installed successfully, open the computer device manager and you will see three devices with yellow exclamation marks in "Other Devices". You can update the drivers in sequence.



PC-side testing method

Note:

Serial port lines need to be cross-connected, that is, the core board's TX is connected to the USB to TTL RX, and the core board's RX is connected to the USB to TTL TX.

The core board's peak power can reach 8W, and the USB power supply capacity of some computers cannot meet this requirement. The core board may restart frequently due to insufficient power supply (manifested by the serial port always outputting power-on information). At this time, it is recommended to use an independent power supply to power the core board. The core board supports 5V~16V power input. When debugging, "common ground" is required, that is, the GND of the core board, power supply, and USB to TTL are connected together.

4. After power on, the core board power indicator (red light) is always on. Wait for about 10s, and the network indicator (blue light) starts to flash. At this time, it means that the core board has been turned on and AT commands can be debugged.

	Indicator status	Module working status
Power indicator light (red light)	light	Power-on
	quench	Poweroff,shutdown
Network indicator light (blue light)	light	Search network status
	200 ms light/200 ms quench	Data connection established, or network registered
	quench	Module shutdown

step	state	AT commands and return values	illustrate
1	send	AT	Handshake test.
	return	OK	If "OK" is returned, it means the serial communication is normal.
2	send	AT+CPIN?	Check SIM card status.
	return	+CPIN: READY OK	If "READY" is returned, it means the card reading is successful. If ERROR is returned, the card reading fails.
3	send	AT+CSQ	Query the RF signal quality.
	return	+CSQ: 24,99 OK	Parameter 1: Signal quality (0~31), make sure the signal quality is greater than 18, otherwise data communication may be unstable. 0 - 113dBm or less 1 - 111dBm 2..30 - 109dBm... -53dBm 31 -51dBm or higher 99 Unknown or undetectable Parameter 2: Not enabled, can be ignored.
4	send	AT+CEREG?	Query network registration status.
	return	+CEREG: 0, 1 OK	Parameter 1: Controls unsolicited result codes, 0 is disabled. Parameter 2: Registration status, 1 indicates successful registration. 0 Unregistered, the module has not currently searched for a new operator to register 1 Successfully registered 2 Unregistered, but the module is searching for a new operator to register 3 Registration denied 4 Unknown 5 Registered, roaming